MESELE TILAHUN BELETE

Postdoctoral Research Associate
Plant virologist and tissue culture expert

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Valid South Korean driver's license

Summary

Biotechnology, tissue culture, and plant virology specialist with extensive expertise in plant virus-host interactions, virus-induced gene silencing, molecular cloning, high-containment plant infection assays, RNA sequencing, cDNA library preparation, and viral population bioinformatics. First author of more than 10 peer-reviewed publications and recipient of the 2024 President's Excellence Award for outstanding Ph.D. research achievement. Experienced in developing plant infectious clones, conducting next-generation sequencing analyses, and designing computational workflows to accelerate plant virology research. Research interests include virus discovery, engineering viral vectors for gene and guide RNA delivery into monocot and cereal crops, and viral diagnosis of economically important crops.

Education

Combined M.Sc.-Ph.D. in Biotechnology, 01/2020 – 08/2024
University of Science and Technology (UST) Daejeon, South Korea

Bachelor of Biotechnology, *University of Gondar* 09/2009 – 07/2012
Gondar, Ethiopia

Experience

Postdoctoral Research Associate, *Animal and Plant Quarantine Agency (APQA)* 09/2024 – Present
Gimcheon, South Korea

- Optimized and streamlined bioinformatics workflows for high-throughput sequencing (HTS), RNA-seq, and virome analyses, reducing data processing time by 20% and enhancing analytical efficiency.
- Developed infectious clone systems and molecular diagnostic assays for plant virus characterization and resistance screening, increasing experimental throughput and data generation capacity by 25%.
- Applied computational and statistical approaches to genomic data analysis, improving reproducibility, accuracy, and interpretability of molecular datasets.
- Designed and conducted high-containment plant infection experiments under Biosafety Level 3 (BSL-3) conditions, supporting plant pathogen research and biosafety risk assessments.
- Processed and analyzed over four HTS datasets and performed more than 3,000 PCR-based diagnostic assays annually for virus detection, identification, and epidemiological surveillance.
- Collaborated with multidisciplinary teams of over 10 researchers, contributing to peer-reviewed publications and the preparation of ongoing manuscripts.
- Oversaw laboratory operations, including equipment maintenance, reagent inventory management, quality control, and compliance with national biosafety and quarantine regulations.

Graduate Research Assistant, 01/2020 – 08/2024
Korea Research Institute of Bioscience and Biotechnology (KRIBB) Daejeon, South Korea

- Performed cDNA library preparation and viral genome annotation for RNA-seq datasets, ensuring accurate genome assembly and functional characterization.
- Conducted RNA-seq and virome analyses across more than 18 HTS datasets representing 105 plant species, accelerating project timelines by 30%.
- Optimized molecular workflows to increase sample-processing throughput by 20%, improving overall laboratory efficiency.
- Generated in-house competent cells for transformation, reducing experimental costs by 10% and supporting routine molecular cloning workflows.
- Authored and co-authored 12 peer-reviewed publications in journals including Virology Journal, Archives of Virology, Plant Disease, and Journal of Plant Pathology.
- Contributed to the design and development of viral vectors for functional genomics and gene silencing applications in plant systems.
- Mentored over 15 graduate students in molecular diagnostics, RNA sequencing, resistance screening, and bioinformatics data interpretation.

Assistant Researcher, *Amhara Regional Agricultural Research Institute (ARARI)* 03/2013 – 01/2020
Bahir Dar, Ethiopia

- Performed over 1,000 PCR-based assays and 30 DAS and NCM-ELISA diagnostics for the detection of plant pathogens in potato and sweet potato, supporting marker-assisted selection in breeding programs.
- Supported virus screening and resistance evaluation in root and tuber crops using ELISA and molecular diagnostic techniques.
- Developed more than 10 optimized protocols for in vitro propagation and germplasm conservation of economically important and horticultural crops using plant tissue culture.
- Conducted large-scale in vitro propagation of crops including potato, sweet potato, ginger, bamboo, and banana, enhancing plant production efficiency.
- Coordinated laboratory setup and decontamination procedures for tissue culture facilities, reducing experimental preparation time by 90%.
- Evaluated germplasm for disease resistance under both field and controlled laboratory conditions, contributing to crop improvement programs.
- Maintained comprehensive and well-documented experimental records to support data integrity, reproducibility, and collaborative research efforts.
- Designed and implemented field experiments and integrated disease management strategies for crop protection.
- Trained more than 100 undergraduate students in molecular techniques and aseptic laboratory procedures, improving technical capacity and laboratory productivity.

Skills

Molecular Biology
PCR, qPCR, cDNA library, ELISA, cloning, transformation, DNA/RNA extraction, vector construction, and gel electrophoresis.

Plant Virology
Virus indcued gene silencing, infectious clones, virus-host interaction assays, greenhouse inoculations.

Plant Tissue culture techniques
Meristem culture, in-vitro mass propagation and optimization protocols

Communication
Multidisciplinary teamwork, collaborative research projects, manuscript preparation, peer review activities, peer-reviewed publications, and international presentations.

Bioinformatics
RNA-seq analysis, sequencing assembly, BLAST, virome profiling, and phylogenetic analysis.

Programming & Data Analysis
R progarme (Intermediate), SAS.

Laboratory Management
Biosafety compliance, equipment handling, inventory control, and BSL-3 facility operations.

Languages

English
Proficient

Korean
Intermediate proficiency

Amharic
Native

AWARDS AND HONORS

President's Excellence Award for outstanding Ph.D. research achievement, 01/11/2023
KRIBB and UST, Daejeon, South Korea

UST Integrated M.Sc. and Ph.D. Scholarship, *UST* 01/11/2019

Recognition Award for in vitro micropropagation of Rosa damascena, 01/11/2011
University of Gondar, Ethiopia

Publications

First report of citrus leaf blotch virus infecting Broussonetia papyrifera (paper mulberry) in South Korea, *VirusDisease.* 15/06/2026
Belete, M.T., Lee, SJ. & Lim, S. First report of citrus leaf blotch virus infecting Broussonetia papyrifera (paper mulberry) in South Korea. VirusDis. (2026).

Towards the Adoption of Genetically Modified Crops in Africa : Navigating Public Perceptions and Biosafety Regulations, 08/01/2025
Korean Biosafety Clearing House (KBCH-biopanel). Belete, M.T.

Report of longan witches' broom-associated virus in lychee imported from China, *Microbiology Resource Announcements* 01/12/2024
Belete, M. T., Park, E., Kim, Y., Yun, S., Lee, J. H., Lee, D. S., ... & Lee, S. J. (2025). Report of longan witches' broom-associated virus in lychee imported from China. Microbiology Resource Announcements, 14(8), e00411-25.

First report of pecan mosaic-associated virus in Atractylodes lancea in South Korea, *Journal of Plant Pathology* 01/12/2024
Belete, M. T., Igori, D., Kim, S. E., Kwon, J. A., Lee, S. H., & Moon, J. S. (2025). First report of pecan mosaic-associated virus in Atractylodes lancea in South Korea. Journal of Plant Pathology, 107(1), 771-772.

First report of cucumber green mottle mosaic virus infecting grapevine in South Korea, *Journal of Plant Pathology* 01/12/2023
Belete, M. T., Kim, S. E., Igori, D., Kwon, J. A., Kim, Y. H., Yang, H. J., & Moon, J. S. (2024). First report of cucumber green mottle mosaic virus infecting grapevine in South Korea. Journal of Plant Pathology, 106(3), 1415-1416.

Molecular characterization and identification of Stellaria aquatica virus C, a novel putative member of the genus Closterovirus infecting Stellaria aquatica, *Archives of Virology* 01/12/2023
Belete, M. T., Kim, D. Y., Kim, S. H., & Sun, M. J. (2024). Molecular characterization and identification of Stellaria aquatica virus C, a novel putative member of the genus Closterovirus infecting Stellaria aquatica in South Korea: MT Belete et al. Archives of Virology, 169(7), 153.

Deciphering the virome of Chunkung (Cnidium officinale) showing dwarfism-like symptoms via a high-throughput sequencing analysis, *Virology Journal* 01/12/2023
Belete, M. T., Kim, S. E., Gudeta, W. F., Igori, D., Kwon, J. A., Lee, S. H., & Moon, J. S. (2024). Deciphering the virome of Chunkung (Cnidium officinale) showing dwarfism-like symptoms via a high-throughput sequencing analysis. Virology Journal, 21(1), 86.

First report of strawberry Polerovirus 1 in strawberry in South Korea, *Plant Disease* 01/12/2023
Belete, M. T., Kim, S. E., Kwon, J. A., Kim, R. H., Kim, Y. H., Yang, H. J., ... & Moon, J. S. (2024). First report of strawberry Polerovirus 1 in strawberry in South Korea. Plant Disease, 108(3), 826.

Complete genome sequence of daphne virus 1, a novel cytorhabdovirus infecting Daphne odora, *Archives of Virology* 16/03/2023
Belete, M. T., Kim, S. E., Igori, D., Ahn, J. K., Seo, H. K., Park, Y. C., & Moon, J. S. (2023). Complete genome sequence of daphne virus 1, a novel cytorhabdovirus infecting Daphne odora. Archives of Virology, 168(5), 141.

Molecular characterization of a novel umbra-like virus from Thuja orientalis (arborvitae) in South Korea, *Archives of Virology* 01/12/2022
Belete, M. T., Kim, S. E., Kwon, J. A., Igori, D., Choi, E. K., Hwang, U. S., ... & Moon, J. S. (2023). Molecular characterization of a novel umbra-like virus from Thuja orientalis (arborvitae) in South Korea. Archives of Virology, 168(7), 197.

Complete genome sequence of pueraria virus A, a new member of the genus Caulimovirus, *Archives of virology* 22/03/2022
Gudeta, W. F., Igori, D., Belete, M. T., Kim, S. E., & Moon, J. S. (2022). Complete genome sequence of pueraria virus A, a new member of the genus Caulimovirus: GW Firomsa et al. Archives of virology, 167(6), 1481-1485.

Complete genome sequence of cnidium closterovirus 1, a novel member of the genus Closterovirus infecting Cnidium officinale, *Archives of virology* 01/12/2021

Gudeta, W. F., Belete, M. T., Igori, D., Kim, S. E., & Moon, J. S. (2022). Complete genome sequence of cnidium closterovirus 1, a novel member of the genus Closterovirus infecting Cnidium officinale: WF Gudeta et al. Archives of virology, 167(6), 1491-1494.

Complete genome sequence of cnidium virus 1, a novel betanucleorhabdovirus infecting Cnidium officinale, *Archives of virology* 01/12/2021

Belete, M. T., Igori, D., Kim, S. E., Lee, S. H., & Moon, J. S. (2022). Complete genome sequence of cnidium virus 1, a novel betanucleorhabdovirus infecting Cnidium officinale. Archives of virology, 167(3), 973-977.

Complete genome sequence of Codonopsis torradovirus A, a novel torradovirus infecting Codonopsis lanceolata in South Korea, *Archives of virology* 01/12/2020

Belete, M. T., Gudeta, W. F., Kim, S. E., Igori, D., & Moon, J. S. (2021). Complete genome sequence of Codonopsis torradovirus A, a novel torradovirus infecting Codonopsis lanceolata in South Korea. Archives of virology, 166(12), 3473-3476.

Certification

Biosafety & Laboratory Safety Training
Korea Research Institute of Bioscience and Biotechnology (2025, 2026).

Molecular Biology & Bioinformatics Training
University of Gondar, Gondar, Ethiopia (2017).

Research ethics training
Hunet, Daejeon, South Korea (2023, 2025).

Training on principles of micropropagation in agrobiotechnology
Better Potato for Better Life (BPBL) Project of CIP, Bahir Dar, Ethiopia (2015).

In-vitro Micro-propagation Training
Workshop at the International Potato Center, Bahir Dar, Ethiopia (2015).

PROFESSIONAL SOCIETY AFFILIATION

BioPanel of the Korea Biosafety Clearing House (KBCH), *Full member* 09/2024

Information center, expert member, responsible for reviewing domestic and international biotechnology papers and policies. Daejeon, South Korea

The Korean Society of Plant Pathology (KSSP), *Full member* 01/2021

References

Jae Sun Moon, Professor, Ph.D., *Professor,*
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Seungmo Lim, Senior researcher, Ph.D., *Senior researcher,*
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Davaajargal Igori, Professor, Ph.D.,, *Professor,*
(Instructor) Mongolian National University of Education, Ulaanbaatar, Mongolia
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Gashaw Belay Alemu, *Ph.D candidate, Graduate research assistant,*
(Former supervisor) University of Wisconsin–Madison, USA
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